Death, Destruction, and Data

An Intro to Using Data to Study International Conflict

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Preface

My name is Miles Williams, and I authored this free and open access book for students taking a 300 level undergraduate class at Denison University called "Death, Destruction, and Data."

This provocatively named course is an introduction to using data to study international conflict. This is a specialized class that assumes some prior knowledge of the R programming language and, ideally, some experience with the `{tidyverse}` dialect. If you don't have prior experience with either, that's not the end of the world. I'll go slowly, and because this class is focused on studying one issue (international conflict) that also means that much of the material will center on conceptual issues rather than an emphasis on technical know-how (though there will be some of that, so don't rest too easy, and I will expect you to do some original data analysis and writing along the way.).

The book is divided into two parts, which I've summarized briefly below.

Part I: Trends in conflict

Part I is all about studying historical trends in international conflict. I begin with an introduction to conflict (including war) and how international relations scholars define armed conflict between countries.

I then talk about how conflict is measured using data, and show how to use this data to answer a pair of important questions: is war becoming more or less common, and is it becoming more or less deadly?

Both questions are foundational to the quantitative study of war, but they have faded in relevance as conflict scholars turned their attention to dyadic (country-pair) analyses of factors that explain why some countries tend to fight while others don't. These questions have witnessed some renewed interest, however, due to recent debates about the "decline of war theory."

On the technical side, we'll spend a lot of time thinking about how conflict is measured, how to visualize trends in conflict onset and deadliness, and detect whether changes in each have taken place.

Part II: Who fights whom and why?

Part II moves the focus to country dyads with the goal of answering the question: why do certain countries fight each other more than others?

The literature on why countries fight is much too broad to cover in a short book about using data to study conflict, but I will draw on a sample of prominent theories of conflict and peace to motivate a variety of analysis examples and expose you to some of the ins and outs of constructing datasets for studying international conflict. I'll also introduce you to the standard set of statistical tools conflict scholars use in their research (so get ready to learn about logit models!).

My hope

I will not be able to address every single technical issue or measurement question that comes along with conflict research. I do hope, however, that these instructional materials will give you a solid enough foundation to start asking your own questions, testing novel hypotheses, and conducting empirical research. This is the start of your journey rather than the end (assuming you choose to keep going long after this course wraps up).

1 What is War?

Main ideas:

- A useful way to think of war is as a continuation of politics by other means (but what does that mean, exactly?).
- Peace science is a research tradition in the field of international relations that seeks to use data and the scientific method to study war and peace.
- War, and international conflict more broadly, is treated by peace scientists as a specific kind of political event that can be measured and, about which, systematic comparisons can be drawn.
- The Correlates of War Project, started in the 1960s, had (and continues to have) outsized influence over how international conflict is defined and empirically operationalized.
- A recent controversy surrounding one of the most popular COW datasets has called into question the reliability of the data, and the outgrowth of this controversy is a brand new dataset that we'll use in the rest of this class.

1.1 Politics by Other Means

It's a bit foolhardy of me to think I can address the question "what is war?" in a single chapter of a set of open source lecture notes. If this were my goal, I'd have to affix a big asterisk to the chapter with the ubiquitous phrase spoken in nearly every *Star Wars* movie ever produced: "I've got a bad feeling about this."

War has been part of the human experience since well before recorded history. For thousands of years, countless thinkers have opined on the nature of war.

However, one of the more recent takes (by historical standards) that you may have heard of is the idea that "war is a continuation of politics by other means." This idea comes from the famed Prussian General, Carl von Clausewitz (2003). Though this sentence is sometimes is taken out of context, and he didn't actually say this exact phrase (he didn't even write in English), this idea that war is just an extension of politics has had a profound impact on contemporary thought about war.

While not all translations or analyses of Clausewitz's writing agree on every detail, few contest the idea that he thought war was inherently political. At least, this is true of the most common type of war. As the political scientist Bear Braumoeller (2013a) nicely summarizes, Clausewitz drew a distinction between two kinds of war: *absolute* war and *real* war.

Absolute war is rare, and considered to be a kind of duel to the death. It also isn't the type of war Clausewitz considered an extension of politics. When he said this, he meant *real* war. So what's the distinction?

As Braumoeller (2013a) further summarizes, real war is a form of political communication meant to gather information necessary to assess the hypothetical outcome of absolute war. Another, more intuitive way to put it, is that real war is about sizing up your opponent. By engaging in brief skirmishes, sides of a conflict learn information about the other side (its strengths, weaknesses, resolve, and so on). This information is necessary so that both sides can agree to the terms of a bargain.

In short, war is intimately connected with bargaining between countries. Countries have some issue, whether that's territory, trade, or resources, that they have a dispute over. As they try to bargain with each other to find a compromise that satisfies them both, there's some uncertainty about how much each side should actually ask for and deserves to get. Real war can serve as a means by which sides get this information. *Ergo*, war as a continuation of politics by other means.

The idea that war is related to bargaining would later be famously formalized by the political scientist James Fearon (1995), who developed a game theoretical model to help him show why countries would choose to go to war when negotiating a deal would be mutually better for both sides. A few years later, R. Harrison Wagner (2000) would extend Fearon's formalized logic of bargaining to include war as not just the result of bargaining failure, but also a costly means by which sides negotiate a deal.

In some ways, these game theoretic approaches nicely encapsulated the ideas Clausewitz proposed long ago. Braumoeller (2013a) puts it most succinctly: "information, not attrition, is the goal of conflict" (9).

Now, if you're at all tuned in to current events, you probably have some doubts about this way of understanding war, and rightly so. You can make a good case that Israel's recent war with Hamas went far beyond communicating information. It seemed an awful lot like attrition. It also demonstrated that war can be a means of decisively changing the balance of power by weakening an opponent, which happened, by extension, with Israeli strikes against Hezbollah in Lebanon and against the Iranian regime.

These kinds of conflicts, however, align more with Clausewitz's definition of absolute war, which he considered rare, as these kinds of conflicts in fact are. But they do happen, and most observers agree that when you initiate a conflict of the continuation-of-politics variety you risk escalation to the duel-to-the-death sort. This is why German Chancellor Theobald von Bethmann-Hollweg referred to the start of war as *rolling the iron dice* on the eve of World War I. Actually, what he is supposed to have said exactly is: "If the iron dice must roll, may God help us."

Iron dice seems an apt analogy, because much to everyone's surprise at the time, the conflict that became known as World War I was predicted by nearly all Great Power elites at the time to be a short conflict that would be over before Christmas (Tuchman 1994). The lesson of World War I is that wars can escalate to attrition, even when those involved had no hopes or intentions of making that happen.

War, in short, is more than just a costly form of bargaining; it comes with existential risks. The choice to roll the iron dice is political, but what happens next can quickly fall beyond anyone's ability to control.

1.2 The Peace Science Tradition

Because war is a form of politics, and of a sort that has fatal implications, it's natural that a group of political scientists would be motivated to carve out a niche area of research focused on trying to study war in a systematic way. This area of research became known as "peace science."

Even up to the mid-20th century, the study of war was mostly the purview of historians, whose analyses generally focused on particular wars or periods in time. The focus was on close examination of historical documents and other source materials to construct a narrative about why a war happened, why one side won or lost, or the special role that particular individuals or social forces played.

These are all important things to document, and the field of peace science would not have been able to get started without the painstaking work of historians to document past conflicts.

However, peace scientists wanted to go further. They wanted to measure wars across time and space in a consistent and rigorous way so that they could then use statistical analysis and the scientific method to systematically test arguments about war and peace. They wanted to come up with theories, generate testable hypotheses, and empirically verify them against observable regularities in conflict.

The political scientist J. David Singer and the historian Melvin Small were among those at the forefront of the peace science tradition. Singer founded the Correlates of War Project in 1963, which remains alive and well today, and with the help of Small they began work to collect data on all wars that took place since 1816 (the post-Napoleonic-Wars period).

On the Correlates of War Project website (most people just refer to it as COW, and pronounce it like "cow"), it says that "[t]he original and continuing goal of the project has been the systematic accumulation of scientific knowledge about war." This project rests on the assumption that international conflicts can be *compared* across time and space. To make this possible, Singer and Small, as the COW website continues, "needed to operationally resolve a number of difficult issues such as what is a 'state' and what precisely is a 'war.'" Based on their work, Singer and Small published a book in 1972 called *The Wages of War*, which would establish a standard definition of war that would guide the work of countless future peace scientists.

As a result of their efforts, decades of research looking at trends in the data and assessing how various factors predict conflict onset have supported the accumulation of knowledge about war. One of the most comprehensive, yet succinct, resources you can find that summarizes all this accumulated knowledge is the edited volume *What Do We Know About War?*, which is now in its third iteration (Mitchell and Vasquez 2024). While there is still a lot that peace science has failed to figure out about war, the field knows a lot more now than they did in 1963 when Singer and Small began their work.

1.3 Defining War and Other forms of Conflict

So how do peace scientists define war? How do they define other instances of conflict between countries as well?

Singer and Small considered war an instance of "sustained combat involving substantial fatalities," according to documentation for the latest COW wars dataset. More precisely, they recorded that a war occurred if more than 1,000 battle deaths were documented, combat was sustained, and it involved organized armed forces.

They would go on to develop a typology of war based on the participants: (1) interstate war, (2) civil war, (3) extra-systemic war.

Interstate war involves wars between and among sovereign countries considered part of the international state system (I'll focus on interstate conflict in this class). Civil war involves conflicts between the government of a state and an armed group within its borders. Extra-systemic war involves conflict between a state and a separate actor not currently recognized as a state, such as a colony.

COW maintains datasets for each of these kinds of wars on their website.

There are other prominent datasets on international conflict as well. Probably the most popular is the Militarized Interstate Dispute (MID, pronounced "mid") dataset, which is also maintained by COW. MIDs are distinct from wars but are often used to study the potential for war, or warlikeness, of countries.

MIDs are a comparatively more recent innovation, and the intended goal of measuring MIDs is to capture more variation in international conflict than is possible using the more stringent criteria used to measure war. The concept and original dataset were proposed by Gochman and Maoz (1984), and the data would go through several rounds of updates and revisions. The most recent version of the dataset (version 5) just dropped in 2020 (Palmer et al. 2022).

COW notes on their webpage that MIDs, citing Jones, Bremer, and Singer (1996):

> '...are united historical cases of conflict in which the threat, display or use of military force short of war by one member state is explicitly directed towards the government, official representatives, official forces, property, or territory of another state. Disputes are composed of incidents that range in intensity from threats to use force to actual combat short of war.'

MIDs, which are not wars, are the most common measure of international conflict used by peace scientists today, and this is for two key reasons.

The first is pragmatic: MIDs are more common than wars. That makes them more amenable to statistical analysis.

The second is conceptual. I think Braumoeller (2013a) puts it best:

> [T]he best way to gauge the warlikeness of nations [is] to measure the frequency with which they run the risk of war. Initiating a war, or issuing a threat that creates the risk of war, is indicative of the willingness, in the worst case, to spill quite a bit of blood; that willingness is the best indicator of the propensity of the state to engage in violent conflict; and the individual conflict incident, or *militarized dispute*, is the most reliable objective indicator of that willingness (10, emphasis mine).

Linking back to the idea of *real* war (war as a costly means of bargaining), MIDs seem to capture the essence of this definition much better than the one proposed by Singer and Small, who require that a war involve actual use of armed military forces and substantial fatalities (at least 1,000). Real war, by Clausewitz's standards, might only need a minimum of a few shots across the bow at sea, or targeted strikes on military bases, for both sides to learn what they need to learn about the other side's ability, resolve, and the possible outcome of absolute war. Threats may even be enough.

Going further, Braumoeller (2019) later concluded in a book he wrote about trends in war that escalation to absolute war (like one of the World Wars) is hard to predict and can happen even when most or all the actors involved wish to avoid it. The time when purposeful political action is the most significant on the path the war is the initial stages when countries issue threats or make the decision to start shooting. MIDs capture this idea by definition. To relate this back to another idea in the last section, MIDs show the willingness of countries to roll the "iron dice."

1.4 How We'll Measure War In This Class

Going forward in this class, we'll primarily use the concept of the MID to study international conflict. However, as I'll discuss more in the next chapter, we won't use the COW MIDs dataset.

Without giving too much away, a recent controversy in peace science emerged in the last decade when a group of scholars undertook painstaking work to validate and recreate the COW MID dataset from scratch. What these scholars found surprised them.

They discovered thousands of data entry errors, including phantom conflicts (conflicts that had no supporting evidence to verify they happened) and double counting. So bad were these issues that this team of researchers claimed all MID data for years prior to 2002 were unreliable for analysis.

This was a bold claim, and a damning one if true. The maintainers of the MID dataset obviously responded, and the result was a series of publications that would slowly come out over several years documenting the argument that ensued. Eventually, the group of scholars who documented these data issues would get a National Science Foundation grant to reconstruct all the data from the ground up, and along the way they made some conceptual and technical improvements.

Perhaps the two most relevant updates are their rigorous reporting of battle fatalities and their choice to redefine wars as a subset of all conflicts rather than a distinct phenomenon. (In the MID dataset, an observation leaves the dataset the moment it escalates to a war.) Their new concept is called the militarized interstate confrontation (or MIC), which can include threats, shows, or uses of force short of war, but also all-out wars.

Even better, these scholars documented specific militarized interstate events (MIEs), their specific timing, and deadliness. These events are the individual threats, uses of force, and battles that comprise MICs.

In short, the data that I'll introduce in the next chapter is incredibly rich and represents an important new advancement in the study of international conflict. I therefore want you to become familiar and comfortable with this dataset as we work with it throughout this class.

1.5 Conclusion

In sum, many in peace science conceptually treat war as political, and in particular as a means of costly bargaining. To measure this concept well, it isn't necessary to only look at massive and deadly conflicts—you can examine threats or even small scale uses of force, too.

Importantly, the field of peace science, which is responsible for providing us with data on international conflicts, bases this exercise on the idea that conflicts are comparable across space and time. By assuming this is true, it's possible to approach the study of war using statistical analysis and the scientific method. And while many questions remain, the field of peace science has been able to use this approach to build up quite a lot of knowledge about war over the decades.

In the next chapter I'll introduce the data that we'll use the rest of the semester, and discus a recent controversy about this data.

2 The Data

Main ideas:

- The Militarized Interstate Disputes (MID) dataset has supported decades of peace science research on international conflict.
- But a dispute over its quality in the last decade spurred the creation of a an alternative set of data: jointly, Militarized Interstate Events (MIEs) and Militarized Interstate Confrontations (MICs).
- We'll use the latter (MIEs in particular) in this course given its purported quality and high level of granular detail.
- Once you download the data, you can see it has lots of useful information, which we'll spend some time exploring.

2.1 Nothing Good Lasts Forever

In the last chapter I talked about war, how it's defined, and how it's measured. I also ended that chapter by noting a controversy that arose about the most commonly used conflict dataset available for studying international conflict: The Militarized Interstate Dispute (MID) dataset maintained by the Correlates of War Project (COW).

MIDs are ubiquitous in conflict research, so when a group of scholars come along and say the data is unreliable for vast stretches of time (every year from 1816 to 2002 to be exact), that creates a stir. It also led to the publication of a series of articles where the competing research teams behind the MID data and the group of discontents duked it out. Many of the articles in the exchange were published in the journal *International Studies Quarterly*, one of the top journals in the field of international relations.

The most significant development to come out of this back-and-forth was the creation of novel datasets by the group of scholars that raised concerns about MIDs. In a pair of articles, the political scientists Douglas Gibler and Steve Miller (2024a, 2024b) introduced the Militarized Interstate Confrontations (MIC) dataset and the Militarized Interstate Events (MIE) dataset. The former is actually an aggregated version of the latter: MIEs are embedded in larger MICs.

The rationale for these new datasets is summarized pretty well in the codebook (a manual about what's in the dataset) for MICs:

> We found an overwhelming number of errors in the original [MID] data and corrected those data, providing all of our suggested changes to the CoWMID Project dataset hosts. However, after more than nine years of communications with CoWMID and several exchanges in *International Studies Quarterly*, it is apparent that CoWMID admits their pre-2002 data is incredibly error-prone but that they have neither the resources nor inclination to fix those thousands of admitted errors. Therefore, we are introducing the Militarized Interstate Confrontation (MICs) data, which are based on proper use of MID coding rules and include numerous advancements in both information and presentation.

While the MIE and MIC datasets include some innovations in the way events are coded, and what information is either included or excluded from the data, both datasets are premised on the foundational ideas behind the concept of MIDs crystallized by Jones, Bremer, and Singer (1996), along with updates made by subsequent iterations of the COW MID project.

In short, Gibler and Miller didn't try to measure a new concept; they tried to measure an old concept better.

That doesn't mean they didn't innovate. One of the most important updates Gibler and Miller made was to keep wars in the data. Recall from the last chapter that MIDs are considered threats, shows, or uses of force between countries *short of war*. Once a conflict escalates to a war, this is noted in the data and then everything from there on out is dropped. Gibler and Miller decided to keep conflicts in the data, even after they escalate to a war. This is why their main aggregate dataset is called the Militarized Interstate *Confrontations* dataset instead of Militarized Interstate *Disputes*. They wanted to signal that they aren't sticking to the original criteria for disputes (events short of war). Instead, they're capturing everything short of war, and all-out-war as well.

I think this is a really useful innovation, because if someone wants to study something closer to the original definition of MIDs, they can do that with this data. And if someone wants to just study all-out-wars based on their own fatality threshold, they can use this data to do that, too.

The way that disputes and wars seamlessly line up in the MIC/MIE datasets is, itself, another helpful innovation. One of the oddest features of the COW MID dataset and the COW Wars dataset is that they can't be easily merged together (ReidSarkees and Wayman 2010). This isn't good. Remember from the last chapter that the COW Wars dataset uses a 1,000 battle death threshold to mark if a dispute escalates to a war. Unfortunately, the COW Wars and COW MID datasets don't totally overlap in their death statistics or even cases they consider conflicts. The MIC and MIE datasets, to the contrary, are unified and completely overlap, and they don't base inclusion on fatality statistics. Instead, they leave it to the researcher to make a judgement call about how to analyze the data, using their own death thresholds or other criteria to study conflicts if that's what they want to do.

I admit that I don't really have a dog in this fight, but for practical reasons I favor the MIE/MIC datasets over the COW MID dataset.

First, I like the consistency of all events, ranging from threats to all-out war, being included in one dataset. That makes my life easier because I don't have to download multiple datasets or toggle between them if I want to study only low-level disputes versus wars.

Second, while I haven't personally validated either the MID or MIE/MIC datasets, I find the case made by Gibler and Miller in support of their data compelling—for two reasons.

For one, as they noted in the quote I shared above, the folks at COW MID admitted that the errors Gibler and Miller found exist, and they said they don't plan to fix them. When people with a vested interest in the validity of a dataset own up to the fact that errors others found were real, that's telling, and then when they say they don't plan to do anything about it, that's frustrating.

Also, while this isn't the most scientifically valid justification, I find it hard to believe that Gibler and Miller would have bothered to get a National Science Foundation grant, put together a research team, and engage in the tedious and lengthy process of reconstructing a dataset that purports to capture *every single* conflict that has taken place between countries from 1816 to 2014 just to win an argument when a good-enough dataset already exists. This seems like the work of sincere researchers who want to make sure fellow peace scientists have the best data possible for doing their work, and who feel like the community of conflict researchers is being let down by current resources.

For these reasons, we will use the Gibler and Miller data in this class. In particular, we'll use the MIE dataset since the MIC dataset is just an aggregated version of its MIE counterpart.

Speaking of which, let me introduce you to the MIE dataset and give you a tour of what it has to offer.

2.2 The Militarized Interstate Events Dataset

First, some background. The Militarized Interstate Events (MIE) dataset and its Militarized Interstate Confrontations (MIC) counterpart were created as a part of the International Conflict Data Project hosted at the University of Alabama. (COW MID also is hosted by a university, previously by the University of Michigan, and now by Penn State.)

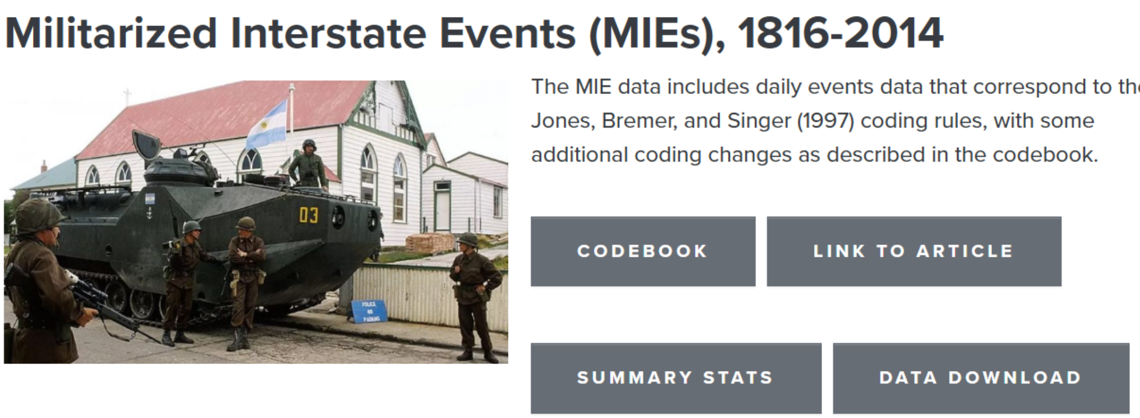
When you go to the download page for the data, you'll see that multiple datasets are hosted on the site (you can check that out here). In addition to the MIE and MIC datasets, there are: Militarized Interstate Participants, Militarized Interstate Confrontation Endings, Militarized

Interstate Confrontation Names (useful if you want to know the historical names of many conflicts that escalated to wars), Protest-Dependent Militarized Interstate Disputes, and Formal War Declarations (which is treated as a separate action from fighting, since declaring war is technically a distinct political behavior from actually going to war).

There's a lot of rich material to draw on here for studying international conflict. Some datasets are just useful summaries of MIE, while others, like the endings dataset or confrontation names dataset provide additional information. For someone like me, it's very easy to spend hours poking around all the files.

But, I have some things to teach you, and you probably want to get a move on, so let me show you the relevant stuff you need to know about the MIE dataset.

First, here's a screenshot of what the download page looks like focusing just on the MIE dataset:



You can see there are four tabs associated with the dataset:

- The codebook (basically, the manual for the dataset)
- A link to the article Gibler and Miller published about the data
- Some bare-bones summary statistics about what's in the data
- A link that will download the data in a zip file to your computer

Don't worry about the last one because I did you a solid and downloaded the data myself and then saved it on my GitHub. This will make it possible for you to read the data into your software (R) directly from online without having to deal with downloading and storing it permanently on your computer.

Before we get to that step though, I think it's a good idea to just look at the information on offer in the dataset. I've copied below the relevant details from the MIE codebook. Each bullet point tells you the name of a column in the dataset and what it represents.

- `micnum`: The Militarized Interstate Confrontation (MIC) number for each case. If the event is part of a Militarized Interstate Dispute (MID) originally coded by CoW, we use that number. Note that we have also added numerous confrontation cases that are not in the CoWMID data as events or disputes. These cases begin with micnum values in the 9000's to demonstrate completely new cases.
- `eventnum`: This is the individual event number within the confrontation. Note that these numbers are meaningful. An event number value of "1" denotes the first threat, display, or use of force in that particular confrontation. The highest event number is the last onset of a threat, display, or use of force that confrontation—previous events may still last longer and cause the duration of the confrontation to extend. A large number of events have missing start and/or end dates. We researched the cases in which the missing-day event could be the first or last participation of the state in the confrontation and confirmed the temporal placement of the missing-day event using a meaningful event number. We have not yet sequentially ordered the missing-day events that could not have been first or last events, but we hope to add that feature soon.
- `ccode1`: This is the CoW country code for the state. ccode2: This is the CoW country code for the state that was targeted by a threat, display, or use of force. In clashes, the higher number ccode is ccode2.
- `stmon`: The start month of the event, with values ranging from 1 to 12. There are no missing month values; if the month of a militarized action could not be determined, the action was not coded.
- `stday`: The start day of the event, with values ranging from 1 to 31 and missing days reported as values of -9. We have done our best to appeal to the historical record to eliminate missing days in the data, but there are numerous cases where only the month of the militarized action is known.
- `styear`: The start year of the event, with values ranging from 1946 to 2014. There are no missing year values; if the year of a militarized action could not be determined, the action was not coded.
- `endmon`: The end month of the event, with values ranging from 1 to 12. There are no missing end month values.
- `endday`: The end day of the event, with values ranging from 1 to 31 and missing days reported as -9. Again, we have done our best to appeal to the historical record to eliminate missing days in the data, but there are numerous cases where only the end month of the militarized action is known.
- `endyear`: The end year of the event, with values ranging from 1946 to 2014.
- `sidea`: Side A in the confrontation is defined by whichever state initiated the first militarized action in the confrontation. All states coordinating with the original Side A state are also considered members of Side A. The variable is dichotomous, with a "1" indicating Side A and "0" indicating Side B. Reciprocated confrontations will have one or more actions by states with a Side A value of "0".
- `action`: The type of action ccode1 targeted ccode2 with in the event. Possible values [with `hostlev` in brackets] include the following: 0 No militarized action [1] 1 Threat to use force [2], 2 Threat to blockade [2], 3 Threat to occupy territory [2], 4 Threat to

declare war [2], 5 Threat to use CBR weapons [2], 6 Threat to join war, 7 Show of force [3], 8 Alert [3], 9 Nuclear alert [3], 10 Mobilization [3], 11 Fortify border [3], 12 Border violation [3], 13 Blockade [4], 14 Occupation of territory [4], 15 Seizure [4], 16 Attack [4], 17 Clash [4], 19 Use of CBR weapons [4], and 22 War Battle [5]. Note that this scale is actually not ordinal. See above for a discussion of how we changed how wars are coded, and see the last section for additional notes on wars and battles.

- `hostlev`: The hostility level of the event. Possible values include the following: 1 No militarized action, 2 Threat to use force, 3 Display of force, 4 Use of force, and 5 War.
- `fatalmin1`: This is an estimate of the minimum number of military battle-deaths for ccode1 in the event.
- `fatalmax1`: This is an estimate of the maximum number of military battle-deaths for ccode1 in the event.
- `fatalmin2`: This is an estimate of the minimum number of military battle-deaths for ccode2 in the event.
- `fatalmax2`: This is an estimate of the maximum number of military battle-deaths for ccode2 in the event.
- `version`: The MIE version number. We are constantly reviewing and updating our datasets, and we ask users of our data to always report which version number of the data they are using in their research.

This is a lot of detail to take in, buut once you have a moment to absorb the information, it boils down to a few key things: identifiers for confrontations and events, identifiers for the country on either side of the event (where the country on side 1 is responsible for engaging in an action against side 2), information about the timing of the event, a classification for what kind of action took place, a classification for how severe the action was, and information about how many people died on each side.

There's enough here that a person can write a lengthy report based solely on what's in this dataset (which you'll do at the end of Part I of this class).

To do that, however, you first need to get the data into your software, and clean it up a bit to make it useful for analysis, which I'll show you how to do in the next section.

2.3 Accessing and Cleaning the Data

Data analysis is impossible without specialized tools, and for the time being the most ubiquitous tool in the political scientist's (and peace scientist's) toolkit is the R programming language and the RStudio IDE (integrated development environment). I'm a political scientist, and I don't drift too far from the herd. I've been using R for more than a decade now.

I can tell you from personal experience that R has its shortcomings, but I'm with the computer scientist Clause Wilke in saying R is among the best programming languages for data science.

More importantly, it's probably one of the best languages for doing statistical analysis and data visualization, which is what most social scientists need to use it for.

Going one step further, as you'll see in Part II of the book, Steve Miller (one of the people responsible for the MIE dataset) created an R package specifically designed to make creating datasets for doing peace science in R easier. This package is going to be a real time-saver when we start incorporating a bunch of different variables into our conflict data to assess which factors predict patterns in international conflict. This package will also help out immensely in the next two chapters when we start visualizing trends in conflict occurrence and deadliness.

But all that in due time. Let me start by showing you how to get the data.

First, you need to make sure you're in RStudio. If your a Denison student, you can access RStudio remotely by going to r.denison.edu, where you can sign in using your Denison credentials. You can also install the software directly to your computer. Just follow these instructions here.

Second, I recommend opening a Quarto file (.qmd) which will give you a single document where you can write in plain text and write and run code. To do this, just go the "File" tab, then "New File," and then select "Quarto Document."

Because my goal in this class is to focus on doing peace science research rather than introducing you to all the basics of using R and navigating RStudio, I'm going to proceed with the naive assumption that you have some background in using these tools.

However, if you don't have this background, all is not lost. Because I have a habit of making textbooks (glorified lecture notes) for all the technical classes I teach, I have written a bunch of guides already about R and RStudio, like this one called "R Basics." If you need some help getting up to speed, I recommend checking this out, and I strongly recommend visiting me during office hours. I'll do my best to coach you up.

One more thing, for those not familiar with R. If you follow along with the code I'll show you for accessing the data and cleaning in up, I think you'll quickly get a sense for how the language works.

2.3.1 Reading the Data into R

The first step in any data analysis task is to access the data you need. In this case, this step is pretty easy, both because a clean dataset already exists, and because I have saved that dataset in a format that makes it possible to read it directly into your computer using some simple R code. I've done so below.

The MIE dataset exists as a CSV file on my GitHub, so I need to give my computer (or if you're using the cloud-based version of RStudio, the server you have remote access to) the location of the file online and instructions to download its contents into its working memory. The below R code allows me to talk to my computer (or the server) to tell it what to do. (This, by the way, is the proper way to think about what R is doing: it's a language that lets you interface directly with your computer, or a remote server, to give it explicit instructions about what to do.)

```
## open {tidyverse}
library(tidyverse)

## read in data and save as 'dt'
read_csv(
  "https://raw.githubusercontent.com/milesdwilliams15/foreign-figures/refs/heads/main/_data/r
) -> dt
```

The first thing I did in the above code is write `library(tidyverse)`. The R language is supported by a vast ecosystem of specialized software packages that allow you to run different kinds of routines (*functions*) in your R session. The `{tidyverse}` package is one of the most popular, and it offers more than just a useful set of functions. It supports an entire dialect in the R language. To install it (if you don't have it already), you can run the following code in the R console: `install.packages("tidyverse")`.

This is the R dialect that I personally speak, and it's become increasingly popular over time. The reason is that speaking R the "tidyverse" way is meant to improve the human readability of code. In short: to help you be nice to the people who need to read your code at a later point in time. I'll use this approach to coding throughout most of the examples in this book.

Once I opened the `{tidyverse}`, I then ran the following code:

```
read_csv(
  "https://raw.githubusercontent.com/milesdwilliams15/foreign-figures/refs/heads/main/_data/r
) -> dt
```

`read_csv()` is a function (a thing that has instructions for a particular routine for your computer to run) that reads a CSV file into your computer's working memory saved at whatever location you tell it to go look. In this case, I gave it the url to the raw CSV file saved on my GitHub. Note that I made sure this url was in quotation marks (" "). This is important for locating files in R.

Note, too, that after I closed the parentheses of the `read_csv` function around the url (and this is also important: make sure you close parentheses), I wrote `-> dt`. In R-speak, this means I assigned the content of the CSV file to be saved as an *object* called `dt`. The arrow is what lets me assign names, and its direction controls what output gets saved with what name. Because the arrow is pointing away from `read_csv()` to `dt`, my computer gets the message that I want it to read a CSV file into its working memory and use "dt" as shorthand for anytime I want to tell it to do something else with this data.

This step is crucial. If I don't save the data as an object, my computer will just read the contents of the CSV file and report them back to me, which isn't helpful. Below, you can see what happens when I don't assign the output to an object and just run the function. Underneath the code you can see a snippet of the output. I told my computer to read in the content of the MIE dataset, and then it just read its contents back to me. Because it isn't saved as an object, I now have no way to refer back to it (except by just re-running the read_csv function every time I need the data, which is just painful).

```
read_csv(
  "https://raw.githubusercontent.com/milesdwilliams15/foreign-figures/refs/heads/main/_data/r
)
```

```
# A tibble: 28,011 x 18
   micnum eventnum ccode1 ccode2 stmon stday styear endmon endday endyear sidea1
    <dbl>    <dbl>  <dbl>  <dbl> <dbl> <dbl>  <dbl>  <dbl>  <dbl>   <dbl>  <dbl>
 1      2        1      2    200     5    -9   1902      5     -9    1902      1
 2      3        1    300    345    10     7   1913     10      7    1913      1
 3      4        2    339    200     5    15   1946      5     15    1946      1
 4      4        3    200    339    10    22   1946     10     22    1946      0
 5      4        4    200    339    10    22   1946     10     22    1946      0
 6      4        5    200    339    10    22   1946     10     22    1946      0
 7      4        6    339    200    10    22   1946     10     22    1946      1
 8      4        7    200    339    10    23   1946     10     23    1946      0
 9      4        8    200    339    11     1   1946     11      1    1946      0
10      4        9    339    200    11     1   1946     11      1    1946      1
# i 28,001 more rows
# i 7 more variables: action <dbl>, hostlev <dbl>, fatalmin1 <dbl>,
#   fatalmax1 <dbl>, fatalmin2 <dbl>, fatalmax2 <dbl>, version <chr>
```

If you're new to the R language, I think the exercise of reading some data into your computer is instructive. There are things called R packages that have useful functions you can use. Functions serve as short-hand for telling your computer to run specialized routines (like accessing data). If you want to save the output of functions, you need to assign their content a name (make it an object, in R-speak), otherwise you'll have to run the function again anytime you need to do anything with its output.

If you can wrap your head around this simple framework, then I think you'll be surprised about how well you pick up coding in the R language.

2.3.2 Cleaning the Data

Compared to many real-life datasets, the MIE dataset is in great shape. First of all, it's **tidy**, which isn't synonymous with "clean." Tidy is a technical term used to describe a dataset where:

1. Each row is one observation.
2. Each column is one variable.
3. Each data entry per row and column is one value.

The MIE dataset ticks these boxes. But what, exactly, is an observation in the dataset?

According to the MIE codebook, each row is a unique militarized event that took place between a pair of countries. The below code will let me look at just the first few rows of data. Take a look at the output.

```
slice_head(dt, n = 5)
```

```
# A tibble: 5 x 18
  micnum eventnum ccode1 ccode2 stmon stday styear endmon endday endyear sidea1
   <dbl>    <dbl>  <dbl>  <dbl> <dbl> <dbl>  <dbl>  <dbl>  <dbl>   <dbl>  <dbl>
1      2        1      2    200     5    -9   1902      5     -9    1902      1
2      3        1    300    345    10     7   1913     10      7    1913      1
3      4        2    339    200     5    15   1946      5     15    1946      1
4      4        3    200    339    10    22   1946     10     22    1946      0
5      4        4    200    339    10    22   1946     10     22    1946      0
# i 7 more variables: action <dbl>, hostlev <dbl>, fatalmin1 <dbl>,
#   fatalmax1 <dbl>, fatalmin2 <dbl>, fatalmax2 <dbl>, version <chr>
```

For each event in the data, there are variables (columns) to help identify the event, indicate which countries were involved, the date it happened, and other details about the kind of military action that took place along with fatality statistics.

Again, in essence, it's tidy: each row is an observation (event), each column is a variable (a different bit of info about the event), and each row-column entry is a single value.

However, even though the data is tidy, it isn't fully ready for analysis. Notice that all the variables are numerical (except the `version` column). For some variables, this is sensible, but not so much for others. For example, the `action` column is meant to be a set of categories for the kind of militarized action that took place, but it's all numbers. The same is true for `hostlev`. These numbers are stand-in numerical codes for categories, and the codebook provides information for what each number represents.

This is a common database management strategy. It's annoying from the perspective of the analyst, but sensible from the perspective of people who store and manage data. Numerical codes eat up less memory than writing out values in plain English. The MIE dataset is 28,011 rows long. This isn't exactly "big data," but it's big enough that some measures needed to be taken to make the overall data file smaller.

The downside for the analyst is that you have to do some cleaning (or "recoding") of variables to make them more useful. Fortunately, in a dataset with only 18 columns, this process isn't too involved. And a nice bonus for you is that I'm about to show you code for cleaning up this data so you don't have to figure it out from scratch. (And down the road I'm going to give you tools that automate this cleaning process even more, so you can focus on the details of your analysis.)

Alright, here's the code. It's a bit more involved than simply reading in data, and I introduced a new kind of operator into the mix called a pipe. This is the `|>` symbol. This is R-speak for "give this thing to that thing." Instead of putting an object inside a function to perform a routine on it, the pipe lets you put the object first. Piping is a way to improve the human readability of code. It also is a way to conveniently chain together a succession of operations you want to perform on your data. The below code tells my computer to convert the numerical codes for `action` and `hostlev` into their qualitative category names. Then it creates a `date` column that is a specialized "date class" variable at the year-month level for when each event started (there are missing values for days, so I'm going to just get as precise as months). Finally, it creates a column called `year`, which is just the event start year (this will help with merging this data with other data down the road). On the tail end of the code I write `-> dt` which saves the output by writing over the old content saved in `dt`.

```
library(socsci)
dt |>
  mutate(
    action = frcode(
      action == 1 ~ "Threat to use force",
      action == 2 ~ "Threat to blockade",
      action == 3 ~ "Threat to occupy territory",
      action == 4 ~ "Threat to declare war",
      action == 5 ~ "Threat to use CBR weapons",
      action == 6 ~ "Threat to join war",
      action == 7 ~ "Show of force",
      action == 8 ~ "Alert",
      action == 9 ~ "Nuclear alert",
      action == 10 ~ "Mobilization",
      action == 11 ~ "Fortify border",
      action == 12 ~ "Border violation",
      action == 13 ~ "Blockade",
      action == 14 ~ "Occupation of territory",
      action == 15 ~ "Seizure",
      action == 16 ~ "Attack",
      action == 17 ~ "Clash",
      action == 19 ~ "Use of CBR weapons",
      action == 22 ~ "War Battle"
```

```
    ),
    hostlev = frcode(
      hostlev == 2 ~ "Threat to use force",
      hostlev == 3 ~ "Display of force",
      hostlev == 4 ~ "Use of force",
      hostlev == 5 ~ "War"
    ),
    date = ym(paste0(styear, "-", stmon)),
    year = styear
  ) -> dt
```

One more thing: I started the code by telling my computer to pull out some tools from a package called `{socsci}`. Remember, packages contain groups of functions (specialized routines). In this case, I want to use a function called `frcode()` to help me turn numerical codes into qualitative categories, and this function isn't already in `{tidyverse}`. To install `{socsci}`, you'll need to run the following code:

```
install.packages("devtools")
devtools::install_github("ryanburge/socsci")
```

2.3.3 Some Simple Descriptive Summaries

Now that I have some cleaned up data, I can use R to tell my computer to do various summaries with the data. For example, I can ask it how many rows are in the data, which will also be a summary of how many unique militarized interstate events (MIEs) took place since each row is a unique event. I can do this by running the function `nrow()` on `dt`. According to the output, the answer is 28,011.

```
nrow(dt)
```

```
[1] 28011
```

I can also ask what time span the data covers. I can do this by running the `range()` function specifically on the `date` variable, which I can pull out of `dt` using a dollar sign (`$`). The first documented MIE took place in July of 1816, and the last happened in December of 2014.

```
range(dt$date)
```

```
[1] "1816-07-01" "2014-12-01"
```

I now can say something really important to know about MIEs. According to the dataset, 28,011 is the number of times that one country threatened to use force, showed force, used force, or did battle in a war with another country between 1816 and 2014 (a nearly 2 century-long period). In absolute terms, that seems like an awful lot of conflict (or the potential for it).

I can also ask my computer (through R) to perform even more detailed summaries. For example, I can ask how common different kinds of military actions took place, and order them by their frequency from highest to lowest. The below code will do just that, and as an added bonus, the `ct()` function that I use below (which comes from `{socsci}`) will also return proportions.

```
dt |>
  ct(action) |>
  arrange(desc(n))
```

```
# A tibble: 17 x 3
   action                          n   pct
   <fct>                       <int> <dbl>
 1 Attack                      11562 0.413
 2 War Battle                   4471 0.16
 3 Show of force                4376 0.156
 4 Clash                        2007 0.072
 5 Border violation             1454 0.052
 6 Threat to use force          1021 0.036
 7 Seizure                       694 0.025
 8 Fortify border                649 0.023
 9 Occupation of territory       643 0.023
10 Alert                         504 0.018
11 Mobilization                  327 0.012
12 Blockade                      148 0.005
13 Threat to declare war          65 0.002
14 Threat to occupy territory     53 0.002
15 Threat to blockade             31 0.001
16 Use of CBR weapons              5 0
17 Threat to use CBR weapons       1 0
```

Now I can say some other interesting things about international conflict. First of all, the most common type of MIE is an attack. From 1816 to 2014 these happened 11,562 times, accounting for more than 41% of all militarized actions by one country against another. The next most common kind of action is battle in an all-out war. From 1816 to 2014, a total of 4,471 war battles took place, accounting for 16% of all militarized actions during this period. Shows of force aren't far behind.

From there the frequency of events declines rapidly. The three least common kinds of actions are threats to blockade, use of CBR (chemical, biological, and radiological) weapons, and the threat to use CBRs. Only 31 blockade threats took place from 1816 to 2014, accounting for 0.1% of all militarized events. CBR use and threats are vanishingly rare. They were only used by one country against another 5 times (short of war) and only one country ever threatened to use them against another. With rounding error, these actions accounted for 0% of all events.

I can also ask my computer questions about the intensity of conflict. Each of the types of military actions is recorded in the MIE dataset as meeting different thresholds for intensity. These intensities are documented in the `hostlev` column in the data. I can run the following code to ask my computer tell me how common these hostility levels are:

```
dt |>
  ct(hostlev) |>
  arrange(desc(n))
```

```
# A tibble: 4 x 3
  hostlev                 n   pct
  <fct>               <int> <dbl>
1 Use of force        15059 0.538
2 Display of force     7310 0.261
3 War                  4471 0.16
4 Threat to use force  1171 0.042
```

Use of force (short of war) is the most common kind of event in the data, accounting for nearly 54% of all military actions by one country against another from 1816 to 2014. Displays of force were the second most common, but occur at only half the rate of uses of force. 7,310 displays of force are recorded in the data, and make up 26% of all events. War battles are even less common, with only 4,471 recorded, accounting for 16% of all events. Threats to use force are the least common. 1,171 in total took place from 1816 to 2014, accounting for only 4% of all events.

I can notch up the complexity of the summaries I ask for yet again. The data also has information about battle deaths. I can write R code that will tell my computer to summarize the minimum and maximum estimated fatalities from different kinds of events based on hostility level:

```
dt |>
  group_by(hostlev) |>
  summarize(
    fatalmin = sum(fatalmin1 + fatalmin2),
    fatalmax = sum(fatalmax1 + fatalmax2)
```

```
  ) |>
  ungroup() |>
  mutate(
    minpct = fatalmin / sum(fatalmin),
    maxpct = fatalmax / sum(fatalmax)
  ) |>
  arrange(desc(fatalmin))
```

```
# A tibble: 4 x 5
  hostlev             fatalmin fatalmax  minpct  maxpct
  <fct>                  <dbl>    <dbl>   <dbl>   <dbl>
1 War                 25737608 31477327 0.994   0.993
2 Use of force          163900   235873 0.00633 0.00744
3 Threat to use force        0        0 0       0
4 Display of force           0        0 0       0
```

This summary offers some really eye-catching headline figures. First of all, wars account for the greatest share of battlefield deaths from MIEs. This fact alone isn't surprising, but the disparity is. A low-end estimate of how many people died in MIEs that were battles in a war from 1816 to 2014 is 25.7 million. The high-end estimate is 31.5 million. Both of those figures in absolute terms are staggering, but they are in relative terms, too. Both the low- and high-end estimates account for more than 99% of all battle-related deaths from international conflicts over a two-century period.

Uses of force short of war can still deadly though. While making up less than a percentage point of all battle-related deaths from MIEs, in absolute terms the totals are still large. On the low-end, 163,900 deaths from 1816 to 2014 can be attributed to uses of force short of war between countries. On the high-end, the figure is 235,873.

Threats and displays of force aren't deadly at all, which isn't a big shock. Neither involves using actual military force. Therefore, they aren't fatal.

2.3.4 Making Some Graphs of Data Summaries

Another thing you can do with your data is make graphs of your summaries. This process is called data visualization, and it involves taking tabular data or data summaries and representing them using geometry, colors, and other graphics.

The R programming language is a great way to interface directly with your computer to get it to produce nice looking graphs. The `{tidyverse}` way to do this is with `{ggplot2}`, which is in the universe of packages supported by `{tidyverse}`. That means you can start using its functions once you've written and run `library(tidyverse)`.

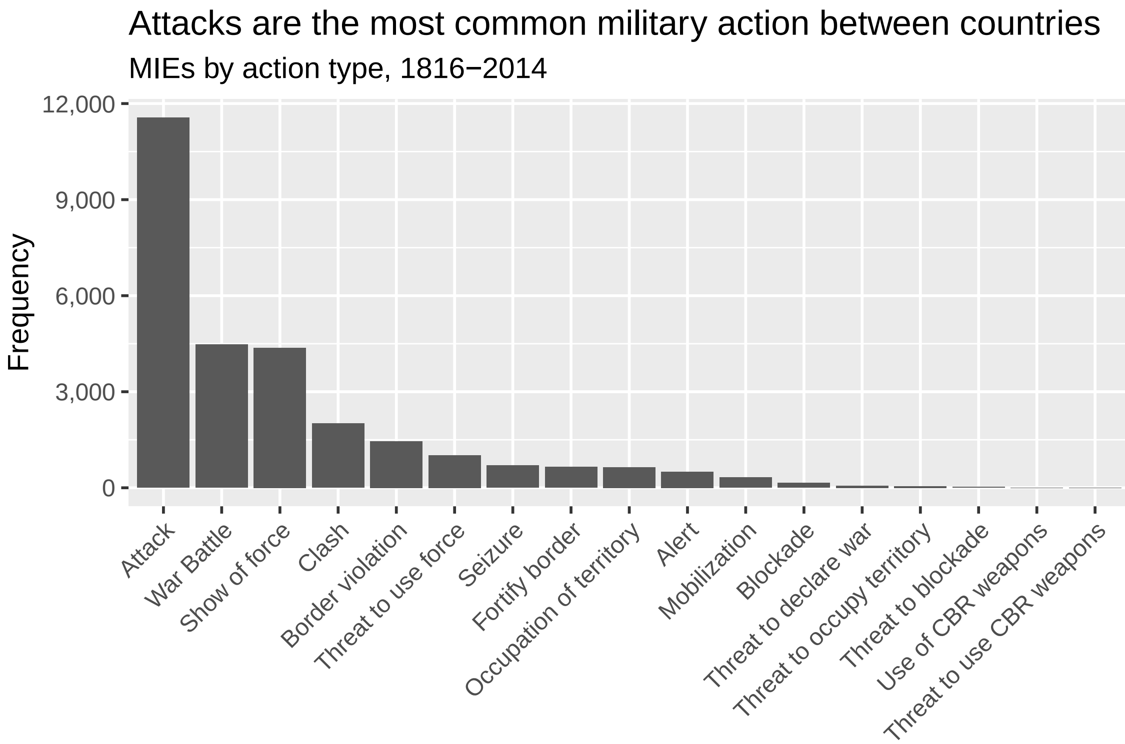
I'm not going to get too bogged down in the details of data visualization here. But if you want to know the whole range of things you can do in terms of data visualization, you can check out some brief lecture notes I've written for my 200-level class on research design, or more detailed notes I've written for a 100-level class I teach on data visualization.

I won't have you do anything too fancy with data visualization in this class, so don't think you need to master everything there is to know. I'll provide you with plenty of examples that you can borrow from and build on throughout this book.

Speaking of examples, it might be nice to have some decent looking graphs of the summaries I showed you in the previous section. Since these examples just involved counting things up, a nice data visualization choice is a simple bar/column plot. These let you summarize how some quantity of interest differs across different qualitative categories.

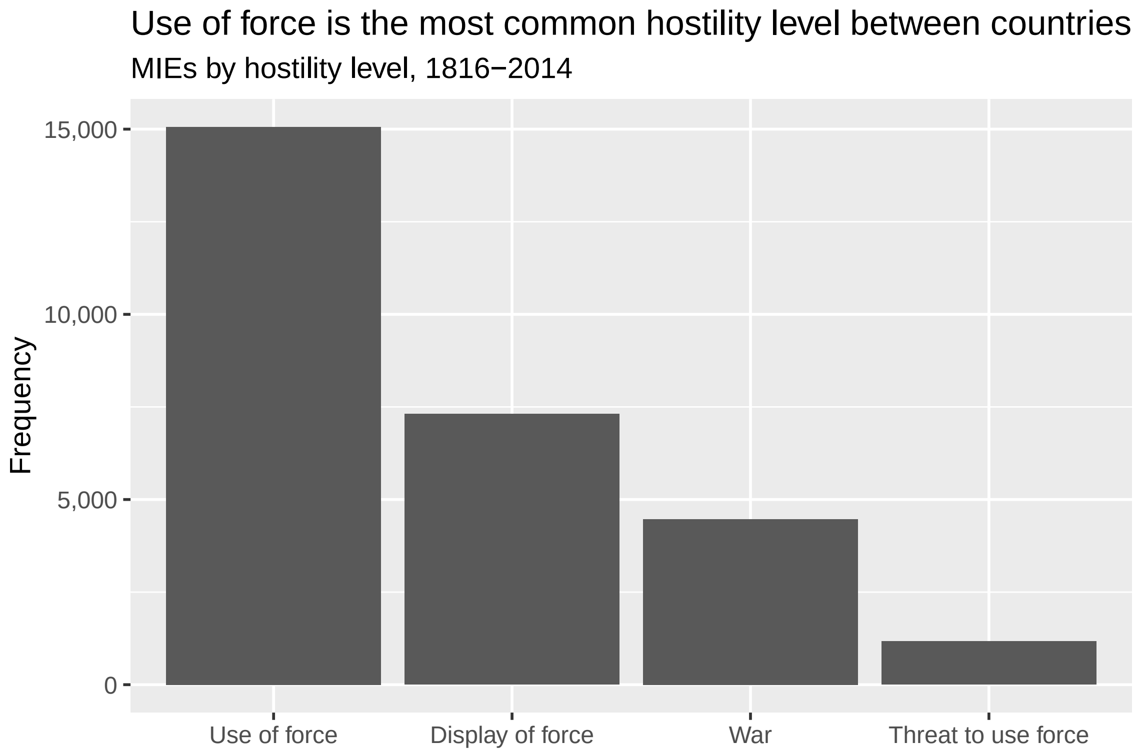
Here's how I might make such a plot using the summary for militarized actions:

```
dt |>
  ct(action) |>
  ggplot() +
  aes(x = reorder(action, -n), y = n) +
  geom_col() +
  labs(
    title = "Attacks are the most common military action between countries",
    subtitle = "MIEs by action type, 1816-2014",
    x = NULL,
    y = "Frequency"
  ) +
  scale_y_continuous(
    labels = scales::comma
  ) +
  theme(
    axis.text.x = element_text(
      angle = 45, hjust = 1
    )
  )
```



Here's another one for hostility level:

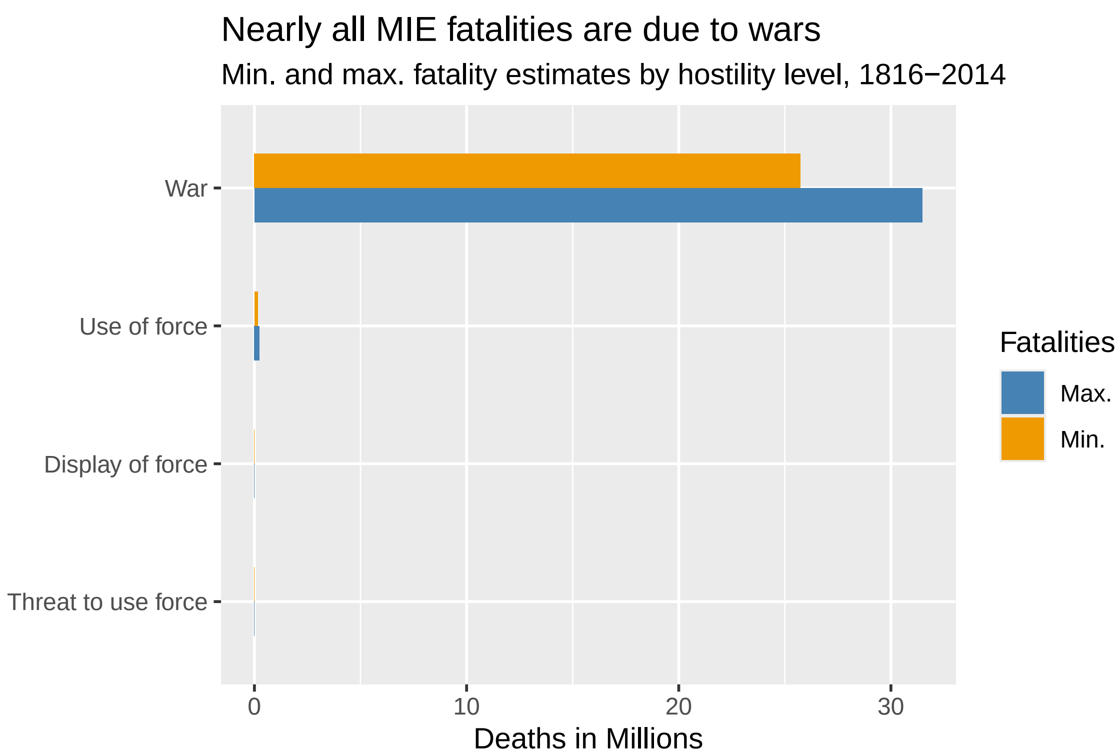
```
dt |>
  ct(hostlev) |>
  ggplot() +
  aes(x = reorder(hostlev, -n), y = n) +
  geom_col() +
  labs(
    title = "Use of force is the most common hostility level between countries",
    subtitle = "MIEs by hostility level, 1816-2014",
    x = NULL,
    y = "Frequency"
  ) +
  scale_y_continuous(
    labels = scales::comma
  )
```



Finally, I can use a bar chart to show fatalities by hostility level:

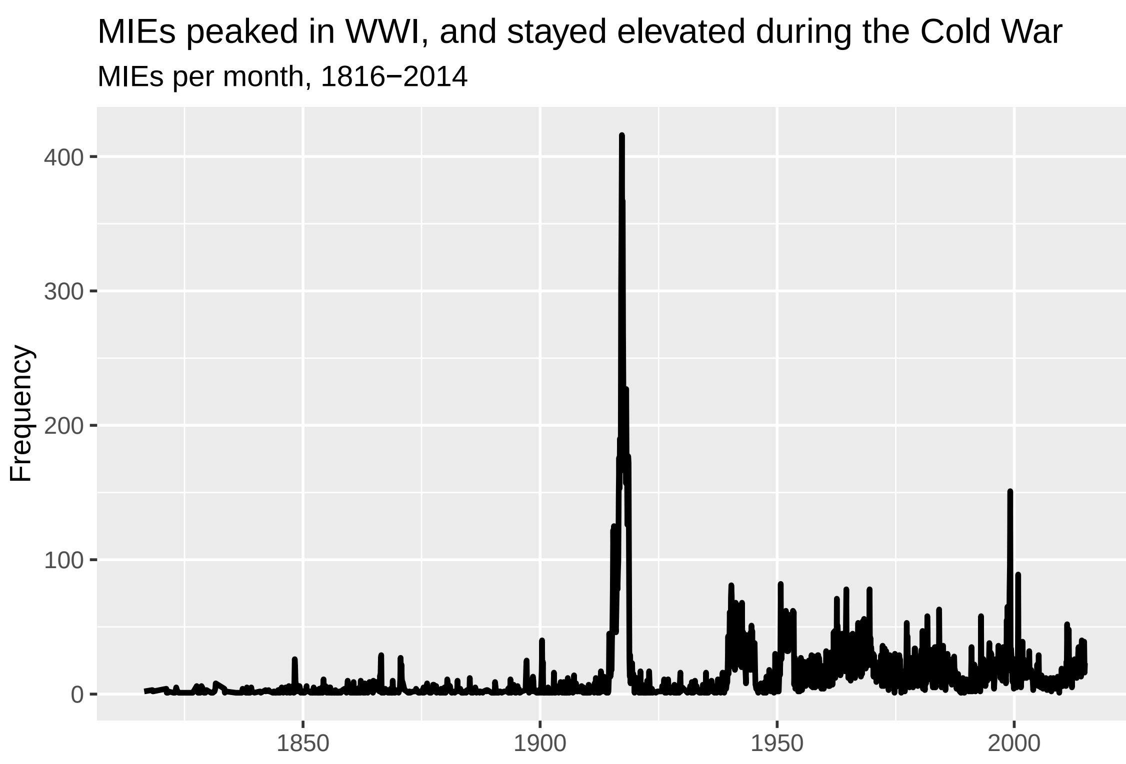
```
dt |>
  group_by(hostlev) |>
  summarize(
    fatalmin = sum(fatalmin1 + fatalmin2),
    fatalmax = sum(fatalmax1 + fatalmax2)
  ) |>
  ungroup() |>
  mutate(
    minpct = fatalmin / sum(fatalmin),
    maxpct = fatalmax / sum(fatalmax)
  ) |>
  pivot_longer(
    fatalmin:fatalmax,
    names_to = "minmax",
    values_to = "fatal"
  ) |>
  ggplot() +
  aes(x = fatal, y = hostlev, fill = minmax) +
  geom_col(
```

```
    width = .5,
    position = "dodge"
  ) +
  labs(
    title = "Nearly all MIE fatalities are due to wars",
    subtitle = "Min. and max. fatality estimates by hostility level, 1816-2014",
    x = "Deaths in Millions",
    y = NULL,
    fill = "Fatalities"
  ) +
  scale_x_continuous(
    labels = ~ scales::comma(.x / 1e06)
  ) +
  scale_fill_manual(
    values = c("steelblue", "orange2"),
    labels = c("Max.", "Min.")
  )
```



One more graph you can put together, which provides a nice segue to the next chapter, is to summarize the number of events by the `date` column in the dataset and then show the results using a line plot. Line plots are a good way to show how some quantity varies over time. The below code shows you one way to do this. The results offer two interesting bits of trivia. First, World War I witnessed the highest number of unique militarized events per month than any other period of time from 1816 to 2014 (far more than even World War II). Second, the monthly frequency of events during the Cold War was elevated compared to previous periods, suggesting a higher baseline level of conflict relative to the previous century.

```
dt |>
  ct(date) |>
  ggplot() +
  aes(x = date, y = n) +
  geom_line(linewidth = 1) +
  labs(
    title = "MIEs peaked in WWI, and stayed elevated during the Cold War",
    subtitle = "MIEs per month, 1816-2014",
    x = NULL,
    y = "Frequency"
  )
```



In the coming chapters I'll talk more about measuring conflict occurrence and deadliness over time, and I'll use ggplot to show these trends. As I said before, I'll also provide you with plenty of examples that you can build on to suit your goals.

2.4 Summary

The Militarized Interstate Events (MIE) dataset is going to serve as the foundation for all subsequent analysis we do in this class. This is a high quality dataset that ostensibly corrects thousands of documented errors in other commonly used datasets by peace scientists (such as the COW MID dataset) and offers some helpful innovations (like keeping events in the data even after escalation to war).

I provided some helpful code for accessing the data, cleaning it, and doing some interesting data summaries. In the chapters that come, I'll provide you with additional resources that help to automate the process of accessing and cleaning the data so you can spend more time thinking about analysis choices and what the data actually says.

3 Is War Disappearing?

Main ideas:

- The decline of war thesis is the argument that conflict is becoming less common and deadly, and that this trend is decades (if not centuries) old.
- A well-known criticism of this position was put forth by the political scientist Bear Braumoeller, first in a well-circulated conference paper (Braumoeller 2013a), and then in a well-received book that fleshed out his criticisms even more (Braumoeller 2019).
- Braumoeller raises many concerns with the decline of war thesis, but two key issues for him were: (1) paying attention to what quantities are counted over time and (2) accounting for the role of chance in war and peace.
- What you count over time concerns not just the numerator (conflict) but the denominator (how you adjust for changes in opportunities for countries to fight).
- Doing statistical inference matters because changes can occur over time that *look* like real changes, when in fact they could just as easily be the result of random chance.

3.1 The Decline of War Thesis & Why It Matters

In this section, I plan to introduce you to the *decline of war thesis*. But before I do, I have a meandering anecdote for you…

I first learned about the decline of war thesis when I read Bear Braumoeller's (2019) book, *Only the Dead: The Persistence of War in the Modern Age*. I was starting my third year as a Ph.D. student, and I learned that Prof. Braumoeller would be visiting my school, the University of Illinois at Urbana-Champaign, to give a talk at the political science department about his book.

I also learned that I'd been volunteered to pick him up from his hotel to take him to campus. I wanted to be prepared when I met Braumoeller for the first time, so I bought and read his book before he flew in from Columbus, OH where he was a professor at *The* Ohio State University. I think I read the whole thing, cover to cover, in just three days (including the R code in the book's appendix for some of the analyses he did).

I devoured the book this quickly for two reasons, neither of which were a pathological need to appear like a smart graduate student to a well-known and respected scholar (or so I tell myself).

First, Braumoeller's previous book, *The Great Powers and the International System: Systemic Theory in Empirical Perspective* (2013b), solidified my desire to study international relations for a living. He published it in 2013, and I read it in 2016 while working on my master's. I had been wondering what would come next once I graduated. Reading Braumoeller's book made my path very clear: I wanted to get my Ph.D. and do what Braumoeller was doing. I've been a fan of his work ever since. So, when I learned that the scholar whose work inspired me to go down the path of getting a Ph.D. to study international relations had just published another book, I couldn't get my hands on it fast enough.

Second, with this new book (much like his first) Braumoeller sought to answer a *big* question in international relations: *is war going away?* Braumoeller had a reputation for doing incredibly smart and cutting edge quantitative work in international relations. If he had applied this same aptitude to one of the most important issues in the field, I had to know what he found.

Needless to say, I had really high expectations going into my first meeting with Braumoeller. That meant I also had a lot of nerves. Because of those nerves, I never directly asked him questions about his book while I drove him from his hotel to UIUC's campus. Also, I thought I'd appear far too creepy expressing how inspiring his work was to me, so I never mentioned it. He asked me some questions about my own research interests, and I think my brain fell out of the back of my head when he did, because I neither sounded eloquent nor coherent as I explained the volatile brew of interconnected research questions percolating in my head.

I kick myself every time I think back on this experience. I met my hero, and I was too star-struck to make the most of it.

So, when I learned that I got a job at Denison, and that the suburb I would live in was the same that Braumoeller lived in (I promise this wasn't on purpose), I thought I could try again. Instead, I got a tough lesson in not letting fear or nerves get in the way of the opportunities life gives you. When I first moved to the area, I learned Braumoeller was overseas on an extended leave doing further research, building on the work he did for *Only the Dead*. And then I learned some tragic news. In the summer of 2023 he unexpectedly passed away while abroad.

I was shocked, as was seemingly the whole field of international relations and even those in political science more generally. I think nearly the whole discipline new him in some way. I also felt incredibly sad, because he had a wife and young daughter that he'd be leaving behind.

I didn't intend to stumble into some "real-talk" when I sat down to write this chapter, but you never truly know the thoughts working themselves out in your mind until you start writing. Consider this detour a nudge to meditate on the value of courage and of taking advantage of the chances life gives you. This nudge comes from a (*still*) young man who's unfortunately already old enough to have accumulated some regrets in life. The experience that fueled this detour (and many others) serve as my own reminders that a little courage will take you a long way.

As off topic as that story may seem, I think it helps me make a key point about studying war.

It's this: studying war with data has an ethical valence to it. Yes, the goal of a peace scientist is to be objective and quantitative about studying war, but because the object of this study is *war*, a life-and-death matter, it should be taken seriously. Above all, I think it requires some *courage* (I knew I could link this back to my anecdote in some way). To make claims about war, using data, is to make claims about the "iron dice" that, when rolled, may lead to no bloodshed and a quick resolution, or to thousands or (in rare cases) millions of deaths. You should make sure that you're serious about the claims you make about war, but you also need to have the courage to make said claims if you think you're right.

In the next chapter I'll talk about how war size can be incredibly volatile and hard to predict, which makes the content of this chapter that deals with the onset of conflict all the more important. If countries are more willing, or less, to roll the iron dice today than in the past, this has existential implications for human lives.

The moral culpability of peace scientists enters the equation when considering how their findings can affect foreign policy and international politics. By making a claim that conflict is becoming rarer, more frequent, or as common as ever, they are making a policy recommendation whether they intend to or not. If the world is getting more peaceful, why should countries continue to invest in international institutions that help prevent conflict? If the world is getting more violent, shouldn't major powers double-down on building up their military might, potentially diverting funds from domestic priorities?

I don't mean to imply that the arc of international relations pivots on the conclusions drawn from a single data analysis. But the goal of data analysis for many social scientists (including those in the peace science tradition) is to make convincing arguments using data. If enough people in a field are convinced of an idea, it can start to catch the attention of practitioners operating in the policy world. Even very niche ideas can attract the attention of a decision-maker. So, while minuscule, a single analysis is part of the body of knowledge and claims accumulated about a subject, and for that reason, it matters.

This, in sum, is why the *decline of war thesis* deserves your attention. It is the argument, founded on a variety of data sources and theoretical propositions, that international conflict is becoming less likely *and* less deadly. Put another way, it's the claim that countries are less apt to roll the iron dice today than decades or even a century ago, and that when the iron dice must roll, they are far less likely to land on another world war. You can imagine that an idea like this catching on can have tangle implications if enough people in the policy world think it's true.

If true, this is great news. But if it's false, it could be a "self-defeating prophesy" (the opposite of its equally devious twin of the self-fulfilling sort). Therefore, getting the decline of war issue right matters.

Braumoeller (2019) disagreed with the decline of war thesis for a variety of reasons, including its self-defeating implications, but in the next section I want to introduce just two of his data-driven critiques, because these will be the basis for the applied examples that follow. Importantly, his data-driven critiques do not on their own disprove the decline of war thesis; they deal with methodological rigor rather than substantive findings. But, Braumoeller claimed that when he applied appropriate measures and methods, the argument in favor of the decline of war thesis looked much less convincing.

3.2 Braumoeller's Critique

Many researchers fall into the decline of war camp, but Braumoeller directs the brunt of his critique at the Harvard Psychologist Steven Pinker who wrote two important books that made their way into the public discourse on war. The first was titled *Better Angels of Our Nature* (Pinker 2012), and the second was titled *Enlightenment Now* (Pinker 2018). In the first, Pinker makes a data-driven case that international war is on the decline and has been for centuries. In the second, Pinker builds on this argument and claims that the Enlightenment movement from the 17th and 18th centuries helps explain why war is going away. He makes the additional argument that the development of modern nation-states also played a role in fostering peace due to their "civilizing" effect on societies.

Braumoeller takes issue with both the data Pinker used, and his theoretical rationales for why war is declining. With respect to the role of the nation-state in promoting peace, Braumoeller notes that the historical record is a mixed bag. On the one hand, when smaller polities organize themselves into larger states, there's more internal peace. On the other hand, states can mobilize for war on a scale that small polities never could. States also have the capacity to commit mass atrocities (think of the Holocaust, or Stalinist USSR).

Braumoeller similarly argues that the Enlightenment is a mixed bag. Some ideas, like valuing reason and respect for rights, are good, but some Enlightenment ideas have borne some rotten fruit. Karl Marx was just as much a product of the Enlightenment as Immanuel Kant, and many atrocities were committed in the name of Marx's ideas, especially in the 20th century. But even seemingly less controversial ideas, like respect for rights, can become a basis for war. The Responsibility to Protect (R2P) was a policy supported by the United Nations (driven in part in response to tragedies like the Rwandan Genocide). Endorsed in 2005, it upholds the idea that if the government of a country is committing human rights violations or mass murder, other countries have a responsibility to intervene, *which* (counterintuitively) *can include using military force.*

But these theoretical complaints about the decline of war thesis aren't the focus of this chapter. Independent of these issues, Braumoeller raises two important methodological concerns about the analysis used to support the decline of war thesis.

The first is that many prior analyses (including Pinker's) measured conflict *frequency* over time, when in fact they should have measured the conflict *rate*. Frequencies are just raw counts, while rates are counts *per* some other quantity of interest. For example, GDP *per* capita is GDP divided by a country's population. For Braumoeller, only counting conflicts over time misses half of the equation. As I'll discuss more in the next section, Braumoeller argues that you need to divide conflict counts by the number of possible chances countries have to fight each other, which doesn't remain constant over time.

The second is that many prior analyses (again, including Pinker's) don't rely heavily, or at all, on statistical inference. Researchers use statistics to quantify the role of random chance in explaining patterns in data. In any variable that goes up and down over time, some of the hills and valleys might indicate a truly new trend in said variable. They could also just be the product of perfectly natural random fluctuations. If you don't account for the role of chance, you might draw overly optimistic (or pessimistic) conclusions from data. I'll discuss this more in the section after the one below.

3.3 What You Count Matters

In the last data analysis example I showed you in the previous chapter, I produced a monthly count of the number of militarized interstate events (MIEs) over time. While this plot provides some interesting fodder for conversation, it doesn't pass Braumoeller's test for a good measure of conflict occurrence. It only shows the frequency of conflict events; not the rate. Further, Braumoeller would probably take issue with counting all the events in the data.

To measure conflict over time to test the decline of war thesis, you need a measure that accurately encodes two important quantities: (1) how often countries are willing to "roll the iron dice," and (2) how often countries have a realistic chance of reaching each other to fight. If you have both quantities, you can then calculate the conflict rate.

It's probably a good idea to express this more formally, which (sorry) means I need to throw a little math your way (it's sums and fractions, so don't sweat it too much). Let me use P to stand for the conflict rate, or what I could also call the *propensity* for war. Further, I'll use I to stand for conflict *initiations*, and O to stand for *opportunities*.

To calculate P for a given point in time t, I need to measure I and O at time t. Theoretically, both are simple to get. I_t is the sum of all conflict initiations at t, and O_t is the sum of all conflict opportunities at t:

$$I_t = \sum_{i=1}^{N} I_{it}, \quad O_t = \sum_{i=1}^{N} O_{it}$$

In the above, I_{it} is an indicator that is 1 if a country decides to roll the iron dice against another; 0 otherwise. O_{it} is between 0 and 1, and it's value represents whether a country can reach another to initiate a fight. The i subscript represents a unique *pair* of countries. In the field of international relations, we call this a *dyad*.

Once you have I_t and O_t, propensity for war is as simple as dividing the former by the latter:

$$P_t = \frac{I_t}{O_t}$$

That math looks temptingly easy, making it dangerous in the hands of someone who doesn't yet know what they're doing. The real challenge is that you, the researcher, need to make some judgment calls about how to get I_t and O_t. Should you count every instance of conflict or only the first instance between a pair of countries? Should threats or shows of force count, or only at minimum when a country decides to use actual military force? How should you measure the opportunity to fight? Maybe by whether countries border each other? But don't some powerful countries have the ability to send their armed forces anywhere in the world? How should you handle that?

Let me give you some possible answers these questions below.

3.3.1 Measuring Incidents

First of all, in his own analysis, which includes his conference paper (2013a) and book (2019), Braumoeller used the Militarized Interstate Disputes (MID) dataset, which at the time his book was published was in its fourth major iteration. Now, I just spent part of the last chapter talking about issues with the MID dataset, which the MIE (Militarized Interstate Events) dataset that we'll use in this book claims to fix. At the time he did his analysis, though, the MID data was the best available. Pinker (2012) used a variety of other datasets in his own analysis to give him longer historical coverage, but many of these have not seen widespread use in the peace science community due to concerns about data quality. Compared to everything available, the MID dataset was best-in-class when Braumoeller used it.

Braumoeller didn't just use the MID data because of its superior quality over other options; he also believed that the *concept* of the MID did the best job of capturing the willingness of countries to roll the iron dice. More specifically, he thought the onset or *initiation* of a MID should be what counts. Anything beyond the initiation of a MID he treated as difficult to predict *ex ante*. Recall that a MID reflects any threat to use force, display of force, or use of force short of war by one country against another. To him this seemed as good a way as any to empirically verify the willingness to risk war. If you recall the discussion in Chapter 1, the definition of a MID comes close to approximating Carl von Clausewitz's definition of "real" war (that is, a continuation of politics by other means).

However, there still is some debate over which MID-like cases should be counted. On the one hand, it makes sense to count everything, including threats and displays of force short of actual violence, because acting threatening is a form of aggression meant to *communicate* resolve and power. And, remember, that use of military force to communicate information about strength and the will to fight is an important part of the Clausewitzian way of understanding war. Such threats can also backfire.

On the other hand, it makes sense to ignore any event short of use of force because use of force is the only action that empirically demonstrates a willingness to fight. Threats and displays of force might just be "cheap talk," but using force reveals that a country will accept bloodshed to achieve its goals. Uses of force also risk reciprocation from the target far more than a threat. If you want to understand the risk of deadly war, you should focus on uses of force where the chance for escalation seems more likely.

Both of these views have merit, which is why Braumoeller (consistent with the practice of many other peace scientists) used the more liberal approach of counting everything, and the more conservative one of ignoring threats and shows of force. The idea is to use both, since each has its merits, and see if it makes a big difference in the results. In many cases, it turns out the this choice actually doesn't matter.

So how can you measure the yearly frequency of conflict incidence using the MIE dataset that I introduced in the last chapter? In short, it's going to require some work to whittle out irrelevant events.

Let me show you a sensible way to proceed. First, I need to get the data, which I do in the below code. Note that the code is a bit different than before. I promised you that I'd provide some tools that make getting and cleaning the data easier, and now I'm delivering on that promise. The below code, like what I showed you before, opens up the packages I need in my R session (`{tidyverse}` and `{socsci}`), but then I use a function called `source()` to which I give a link for a .R file on my GitHub. This is an R "script" file (essentially one giant code block saved as a separate file). This script contains code that creates a function called `get_mie_data()` that performs a straightforward routine when I use it: it reads in the MIE .csv file from my GitHub and then implements the recodes I showed you in the last chapter before returning the data. This is a major time-saver and makes my code easier to read. After running the `source()` function on my R script I now can use the function called `get_mie_data()` and then save the output as an object (in this case, one called `dt`).

```
## set up
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.2.3
```

```
Warning: package 'ggplot2' was built under R version 4.2.3

Warning: package 'tibble' was built under R version 4.2.3

Warning: package 'tidyr' was built under R version 4.2.3

Warning: package 'readr' was built under R version 4.2.3

Warning: package 'purrr' was built under R version 4.2.3

Warning: package 'dplyr' was built under R version 4.2.3

Warning: package 'stringr' was built under R version 4.2.3

Warning: package 'forcats' was built under R version 4.2.3

Warning: package 'lubridate' was built under R version 4.2.3

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.2     v readr     2.1.4
v forcats   1.0.0     v stringr   1.5.0
v ggplot2   3.5.0     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.0
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become e
```

```r
library(socsci)
source("https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/

## get the cleaned up data
get_mie_data() -> dt
```

```
Rows: 28011 Columns: 18
-- Column specification ------------------------------------------------------
Delimiter: ","
chr  (1): version
dbl (17): micnum, eventnum, ccode1, ccode2, stmon, stday, styear, endmon, en...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Now that I have the data, I can start working on a measure of conflict incidents over time. While the data has a `date` variable that documents the unique month and year a conflict starts, most conflict research aggregates data to the year, so I'll do the same. This approach has some practical advantages, like the fact that most variables people want to use to study conflict are only at the year level, and the fact that with nearly 200 years worth of data, monthly variation isn't quite as important as yearly variation.

But what, exactly, should I count per year? To capture what Braumoeller was after, I need to count up two quantities: (1) the total number of unique conflicts documented in the data—not *events*—and (2) the total number of unique conflicts that at minimum rise to the level of a use of force. Remember that the MIE dataset documents each an every event associated with a conflict (or *confrontation*, to use their terminology). This makes for a really rich dataset, but for the purpose of simply measuring war propensity, I need to cut some fat.

The below code lets me extract the relevant quantities from the data. I start by grouping the data by `micnum` (the unique confrontation, or MIC, identifier). I then get the first year for which the MIC appeared, and the maximum hostility level observed. Next, I have to group the data by year to get a count of the unique number of MICs that occurred per year, and the sum of MICs that at least rose to the level of use of force. Finally, I use the `complete()` function to fill in gaps in the data (years that didn't witness any MICs) with zero values. I then save the output as a new object called `incident_dt`. I annotated the code so you can see where each step takes place.

```
dt |>
  # for each MIC get the start year and max hostlev
  group_by(micnum) |>
  summarize(
    year = min(year),
    hostlev = max(hostlev)
  ) |>
  # for each year get total MICs and only MICs with >= use of force
  group_by(year) |>
  summarize(
    incidents_any = n(),
```

```
    incidents_force = sum(hostlev >= "Use of force")
  ) |>
  # complete the series by filling in years without MICs with zeros
  complete(
    year = full_seq(year, 1),
    fill = list(incidents_any = 0, incidents_force = 0)
  ) -> incident_dt # save as incident_dt
```

If you print out the first 10 rows of the data, you can see that I now have a simple dataset. Each row is a year from 1816 to 2014, and for each year I have two counts of conflict incidents—one that counts every conflict regardless of its hostility level, and another that counts only conflicts that at minimum involved use of force. In short, I have everything I need for the numerator in the equation for war *propensity*, but that still leaves the other half of the equation.

```
slice_head(incident_dt, n = 10)
```

```
# A tibble: 10 x 3
    year incidents_any incidents_force
   <dbl>         <int>           <int>
 1  1816             1               1
 2  1817             0               0
 3  1818             1               1
 4  1819             0               0
 5  1820             0               0
 6  1821             4               2
 7  1822             1               1
 8  1823             0               0
 9  1824             0               0
10  1825             0               0
```

3.3.2 Measuring Opportunities

To accurately measure war propensity, a simple count of yearly conflict won't do. I need to know how common new conflicts are relative to the opportunities that exist for new conflicts to arise.

The problem (you knew there'd have to be a problem, right?) is that there isn't one universally accepted definition of what counts as an opportunity. However, the basket of options is, thankfully, limited.

The first option is to naively assume that all countries in the world can reach each other with equal ease, no matter geography or ability. If you want to make this assumption, your measure of opportunity is straightforward: just count up the total number of unique country pairs (*dyads*) that exist in the world at a given point in time. Put another way, assume that for a single country, like the United States, any other country in the world is fair game. This means that the U.S. has as many opportunities to go to war as there are countries in the world. Multiply that total by a count of all countries in the world and there's your measure of opportunity.

You can probably see that this approach has some conceptual problems, however. For argument's sake, assume that the U.S. really can fight any other country it wants. Can you say the same of Zimbabwe? Or Jamaica? What about Switzerland? It seems silly to assume that any other country in the world is fair game for every country in the world.

This takes me to the second option, which is on the opposite end of the spectrum. One idea, which Braumoeller (2013a) explored in his conference paper, is to only count country pairs or dyads that are *contiguous*. This term just means that countries border each other (though some define it more broadly to include countries that are separated by no more than 400 miles of water). This definition of an opportunity to fight rules out the vast majority of other countries in the world for each individual country.

While this approach seems sensible for some cases, it definitely doesn't work in every case. For example, this approach would say that the U.S. has only three opportunities for war: Mexico, Canada, and Russia (Alaska is only separated from it by 50 miles of water). It should go without saying the U.S. has many more opportunities for war, as recent events attest.

There is, thankfully, a middle ground between counting everything and only counting contiguous country pairs. The solution that most peace scientists use today involves a compromise: for some countries, only contiguity applies, while for others, any other country is on the table. The countries for whom every other country in the world is a possible target are called "major powers." These are countries that, in theory, have sufficiently powerful armed forces and resources to project their military might anywhere in the world. At the moment, this list includes countries the United States, China, and Russia. This measure is known as *politically relevant dyads* (PRDs).

Admittedly, this solution also has some issues because most sensible people would agree that the U.S. has very different capabilities than Russia. Nonetheless, it's probably better than the alternatives. However, there is another option, which, funnily enough, also comes from Braumoeller along with his colleague Austin Carson (2011).

Their approach bases opportunities on quantities generated from a sophisticated statistical model that uses major power status, contiguity, and the distance between countries to quantify the probability that one country could fight another (assuming it wanted to). The difference with this measure compared with the alternatives is that opportunities are probabilistic rather than deterministic. It also means that even geography can be a limiting factor for major powers. While the U.S. can certainly send its forces anywhere it wants, it is logistically harder to do so in Iran than in Venezuela. To keep things concise, I'll just call this measure of opportunities *Braumoeller-Carson dyads* (or BCDs).

There are issues with this measure, too. For instance, it still treats major powers equally, and it assumes that distance means the same thing to the U.S. that it does to Bolivia. However, it's probably a stronger option than the other three, but given the prevalent use of the the previous one (PRDs), it's wise to use both it and the Braumoeller-Carson measure to see if using one or the other makes a big difference.

This hard to do without data, of course, so with some measures of opportunities identified (PRDs and BCDs) the next step is to actually get some data. Thankfully, because of the hard work of the peace science community, the relevant data and tools to access it only require writing a few lines of R code. I also did some leg work by creating another R script file that gives you a pre-canned routine for generating the data based on these other tools. All you need to do is run the `source()` function on the url address for the relevant script saved on my GitHub, and then you can use the function `get_opportunity_data()` to access yearly counts of conflict opportunities.

```
source("https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/

get_opportunity_data() -> opp_dt
```

```
Joining with `by = join_by(ccode1, ccode2, year)`
```

If you take a peek at the object `opp_dt` created by the above code, you can see what's in the data. For each year, it counts up opportunities using all four of the methods outlined above (I wanted to be comprehensive, even though I don't recommend using all four). By looking across the rows you can see that these methods give very different values. As you'd expect, `dyads` (which counts every possible pairing per country) is the most liberal count, while `contdyads` (which only counts contiguous pairings per country) is the most conservative. Both the PRD and BCD measures are in the middle, with the latter a bit more conservative, which isn't surprising since BCD is probabilistic rather than deterministic, and it also accounts for the distance between countries. For this reason, the BCD sum per year isn't a whole number. Unlike the other measures, a pair of countries isn't a 0 or 1, but something in between.

```
slice_head(opp_dt, n = 10)
```

```
# A tibble: 10 x 5
    year dyads contdyads   prd   bcd
   <dbl> <int>     <dbl> <dbl> <dbl>
 1  1816   506       120   248  199.
 2  1817   506       120   248  199.
 3  1818   506       120   248  199.
 4  1819   506       120   248  199.
```

```
 5  1820    506        120    248   199.
 6  1821    506        120    248   199.
 7  1822    552        120    258   205.
 8  1823    552        120    258   205.
 9  1824    552        120    258   205.
10  1825    600        126    272   215.
```

3.3.3 Measuring War Propensity

Now that I've made a dataset of new conflict incidents per year, and another of conflict opportunities per year, I need to combine them together so that I can construct a measure of war propensity. This step is easy, because propensity is just the sum of incidents divided by the sum of opportunities.

I do, however, have alternative measures for these different concepts. I have two ways of quantifying incidents, and another two sensible ways of quantifying opportunities. That gives me four alternative measures of war propensity.

To create all four, I first need to combine the incidents and opportunity datasets together, which I can do easily with tools from the `{tidyverse}`. The below code uses a function called `full_join()` that joins the datasets together based on matches in the `year` column in each:

```
full_join(
  incident_dt,
  opp_dt,
  by = "year"
) -> final_dt
```

I saved the combined data as an object called `final_dt`. If I print out the first 10 rows, I can confirm that the datasets merged together as intended:

```
slice_head(final_dt, n = 10)
```

```
# A tibble: 10 x 7
    year incidents_any incidents_force dyads contdyads   prd   bcd
   <dbl>         <int>           <int> <int>     <dbl> <dbl> <dbl>
 1  1816             1               1   506       120   248  199.
 2  1817             0               0   506       120   248  199.
 3  1818             1               1   506       120   248  199.
 4  1819             0               0   506       120   248  199.
 5  1820             0               0   506       120   248  199.
 6  1821             4               2   506       120   248  199.
```

```
 7  1822               1                  1   552         120    258   205.
 8  1823               0                  0   552         120    258   205.
 9  1824               0                  0   552         120    258   205.
10  1825               0                  0   600         126    272   215.
```

The next step is to modify the data by including columns for the four possible measures of war propensity. You can do this with the `mutate()` function, as I do below. This code will add four new columns to the dataset called `prop1` up to `prop4` that are made by taking one of the incidents measures and dividing by one of the opportunity measures. Now, you have all the data you need.

```r
final_dt |>
  mutate(
    prop1 = incidents_any / prd,
    prop2 = incidents_force / prd,
    prop3 = incidents_any / bcd,
    prop4 = incidents_force / bcd
  ) -> final_dt
```

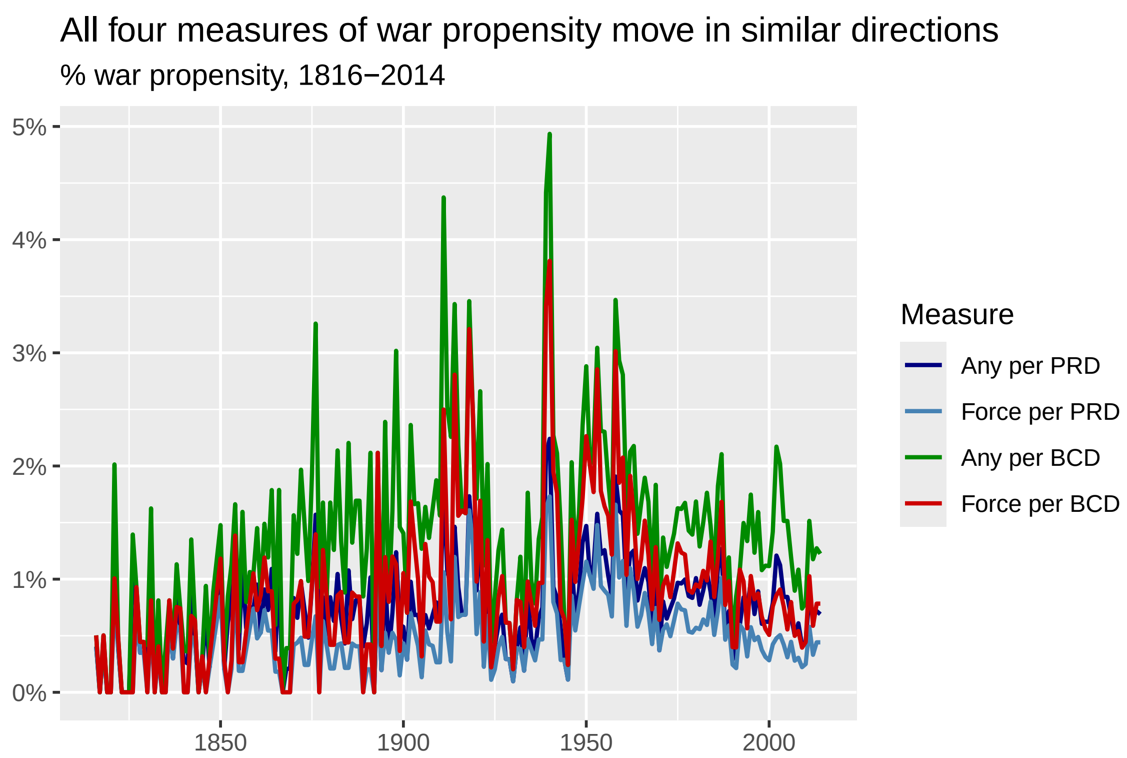
3.3.4 Looking at the Trend

At long last, we have some data to measure variation in war propensity over time. So let's take a look at the results.

The below code will create a line plot showing, per year from 1816 to 2014, war propensity based on each of the four measures created in the previous section. As you can see, the measures aren't identical, but they do move in parallel fairly well, which tells me the choice of measure may not have too much of an impact on the conclusions you draw about conflict over time.

```r
final_dt |>
  pivot_longer(prop1:prop4) |>
  ggplot() +
  aes(x = year, y = value, color = name) +
  geom_line(linewidth = .75) +
  scale_color_manual(
    values = c("navy", "steelblue", "green4", "red3"),
    labels = c("Any per PRD", "Force per PRD", "Any per BCD", "Force per BCD")
  ) +
  scale_y_continuous(
    labels = scales::percent
  ) +
```

```r
  labs(
    title = "All four measures of war propensity move in similar directions",
    subtitle = "% war propensity, 1816-2014",
    x = NULL,
    y = NULL,
    color = "Measure"
  )
```



This just leaves us with with one final question: does the data support the decline of war thesis? Stated differently: are conflicts becoming less common over time?

One way to answer this question is to simply look at the results in the above figure. This is what folks like Steven Pinker did, so why not do the same? However, as I'll discuss in the next section, looking at the data is too subjective an approach. It's much better to rely on statistical inference to help separate random noise from true signals in the data.

3.4 Accounting for Random Chance Matters

Getting the right data and the right way to measure conflict over time is essential for testing the decline of war thesis, but for Braumoeller, this doesn't go far enough. His other methodological critique of past work on the decline of war is the failure to account for the role of chance in war and peace. Statistical analysis can help solve this problem.

There are many ways of defining statistics, but I think the most helpful is to think of it as a set of tools for reasoning about, and quantifying, the role of random chance in generating data.

Imagine that when you measure something in the world, the final number that you record is the product of two things: (1) a signal and (2) some background noise. Statistics provides a way to quantify on average how much the recorded estimate might be expected to vary from one case to the next due to background noise.

Beyond quantifying noise, when doing statistical analysis, researchers will apply certain thresholds of random variation to judge whether background noise can be rejected as the best explanation for a change or difference observed in data. These thresholds are called the *level of the test*, and they are pivotal to the practice of *hypothesis testing*. The idea is simple:

1. Start from the assumption that differences or changes you observe in data are the result of background noise.
2. Estimate a quantity, or quantities, of interest (a difference, a change, maybe multiple estimates).
3. Caculate how much you expect the quantity of interest to vary due to random noise.
4. Reject the hypothesis that background noise explains the estimates you observe if a difference is bigger than a pre-specified threshold of random variation.

If you've never taken a statistics course before, don't worry. R will do all the math for you. If you can at least wrap your head around the point of doing statistical analysis, that will take you a long way.

Let me help you out by showing you a quick example with the conflict dataset I put together in the last section (`final_dt`). Say I wanted to test whether the average conflict rate from 1820 to 1849 is statistically the same as the average conflict rate from 1850 to 1879. I would need to first calculate the observed average conflict rate in each of those periods, then I would need to use statistical tools to calculate intervals around those averages that capture how much variation I would see in the estimated average due to background noise. I can set the level of these intervals based on how high I want the threshold to be for ruling out background noise as the best explanation for any differences I observe between periods.

The below code shows how I might go about doing this using the first war propensity measure (all new confrontations per PRDs). As I told you, I'll provide a lot of tools to streamline the process of doing analysis, and this example is no exception. It starts by reading in a script file that creates a function called `boot_mean_ci()`, which does two things. First it calculates a mean for some numerical variable, then it computes intervals for that mean that reflect how much it might vary as a result of background noise. The method it uses to do this is called bootstrapping, which is a computational method for simulating the distribution of some estimate as a result of random noise. It then uses this distribution to calculate upper and lower bounds for *confidence intervals*. By using the `ci =` option I can control how wide I want these intervals to be. In this case, I set them to .84 (on a scale from 0 to 1) which means it will return 84% confidence intervals. Intervals of this size are the most common level chosen for checking if two quantities are different from each other based on whether the intervals for the quantities overlap. (There's another interval that you might be more familiar with, the 95% confidence interval, that will be relevant in the second part of this book).

```
source("https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/

final_dt |>
  filter(year %in% 1820:1879) |>
  group_by(year < 1850) |>
  summarize(
    mean_ci_boot(prop1, ci = .84)
  )
```

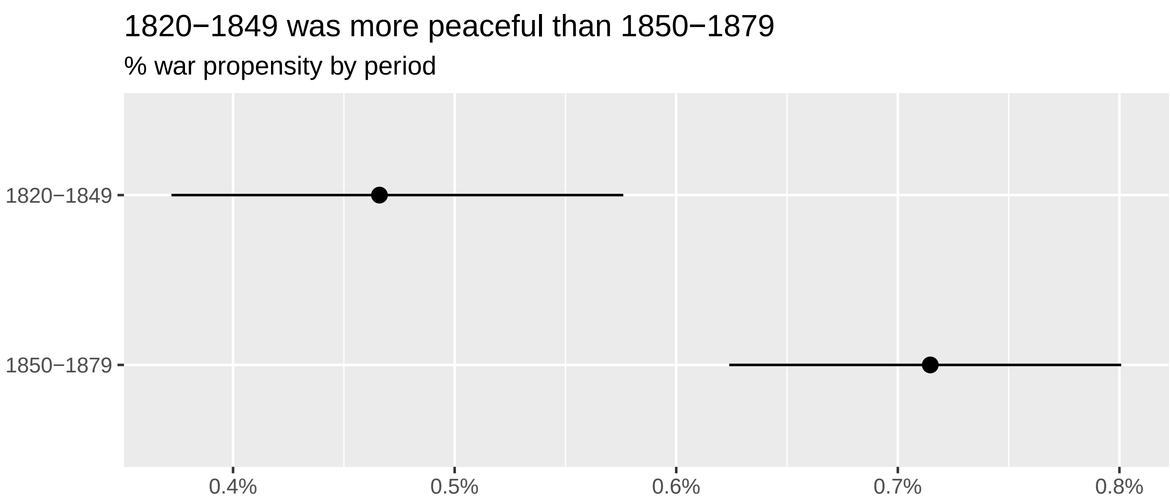
```
# A tibble: 2 x 4
  `year < 1850`    mean   lower   upper
  <lgl>           <dbl>   <dbl>   <dbl>
1 FALSE         0.00715 0.00630 0.00807
2 TRUE          0.00466 0.00368 0.00573
```

If you look at the results, it seems that the average conflict rate from 1820 to 1849 was *lower* than the rate from 1850 to 1979, and that this difference is *statistically significant*. You can tell because the upper bound for the confidence interval for the pre-1850 average is less than the lower bound of the post-1950 average.

This is much easier to see by getting R to produce a figure based on the results, which the below code does:

```
final_dt |>
  filter(year %in% 1820:1879) |>
  group_by(year < 1850) |>
  summarize(
    mean_ci_boot(prop1, ci = .84)
  ) |>
  ggplot() +
```

```
  aes(x = mean, xmin = lower, xmax = upper, y = `year < 1850`) +
  geom_pointrange() +
  scale_x_continuous(
    labels = scales::percent
  ) +
  scale_y_discrete(
    labels = c("1850-1879", "1820-1849")
  ) +
  labs(
    title = "1820-1849 was more peaceful than 1850-1879",
    subtitle = "% war propensity by period",
    x = NULL,
    y = NULL
  )
```



To test the idea more generally that conflict over time has changed in an upward or downward direction, all you have to do is take the concept applied above and extend it to different periods in time, which is essentially what Braumoeller did in his own analysis; though, with a slightly different method. However, to keep things simple, let's just extend the above analysis to include the whole range of years in the dataset.

The idea with the above example is to bin the data into discrete periods in time and then see if the average of war propensity in one period is statistically different than what it is in another. You can use bigger or smaller bins, which will change the results somewhat. Obviously, smaller bins will show more variation from one point in time to the next (but they also will be subject to more background noise). Conversely, bigger bins will give you more stable results (and be less subject to background noise), but this will come at the cost of glossing over potentially important and meaningful variation across shorter time intervals.

Let me show you an example that breaks it down by 20 year intervals. I left some annotations in the code so you can see what is going on in the different steps. Essentially, it lumps the data into 20-year intervals and creates some sensible looking labels for each period. Then it calculates the average war propensity per period along with 84% confidence intervals using the first measure of war propensity (any new confrontation divided by PRDs). Finally, it gives the results to `ggplot()` for plotting.

```
final_dt |>
  # add numerical labels for each 20 year period
  mutate(
    period = round(1:n() / 20) * 20
  ) |>
  # add labels that specify the year range per period
  group_by(period) |>
  mutate(
    period_label = paste0(min(year), "-", max(year))
  ) |>
  # make sure the labels are treated as an ordered category
  ungroup() |>
  mutate(
    period_label = factor(
      period,
      levels = period,
      labels = period_label
    )
  ) |>
  # get the mean and CIs by period label
  group_by(period_label) |>
  summarize(
    mean_ci_boot(prop1, ci = .84)
  ) |>
  # visualize the results
  ggplot() +
  aes(
    x = period_label,
    y = mean,
    ymin = lower,
    ymax = upper
  ) +
  geom_pointrange() +
  scale_y_continuous(
    labels = scales::percent
  ) +
```

```
  labs(
    title = "War propensity peaked in the mid-20th century, but propensity today is about\nas
    subtitle = "% war propensity by 20 year increments",
    x = NULL,
    y = NULL
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

The results are consistent with a trend that is a bit more complicated than the one proposed by the decline of war thesis. It looks like the propensity for war increased between the early and mid to late 19th century. It then increased again in the early 20th century (likely due to WW1) before briefly returning to mid to late 19th century levels. There's a spike again in the mid 20th century due to WW2 and then a decline, yet again, to mid to late 19th century levels. One sensible conclusion you could draw from the data is that the most peaceful time in international politics was the early 19th century, but since then (aside from upswings during the world wars), war propensity tends to return to a stable rate of a little less than 0.75%, which was the norm for most of the 19th century and beginning of the 20th, the inter-war period between WW1 and WW2, and the early 21st century. It was slightly elevated during the Cold War, but the trend during this period just looks like a gradual decline back to the norm.

In short, the typical conflict rate over the past two centuries has remained mostly stable, with the exception of the world wars, and the unusually peaceful period in the decades after the Napoleonic Wars. Braumoeller drew about the same conclusion using an older version of the MID dataset.

This of course leaves open the question of how much the results change if you use one of the other measures of war propensity I included in the data. But I need to give you something to investigate on your own, so I'll leave it to you to figure out how fragile the findings are to the choice of measure.

3.5 Summary

In sum, Braumoeller leveled two methodological critiques on past efforts to test the decline of war thesis: (1) ill-conceived measures and (2) the failure to account for chance.

In this chapter, I covered some of the conceptual hurdles that attend measurement, and showed you how, once you have some sensible measures, you can account for the role of random chance in explaining patterns in the data.

With respect to an appropriate measure of conflict over time, the best approach is to calculate the frequency of new conflicts *per* the number of opportunities for new conflicts to arise. The overall conflict count can go up and down for a variety of reasons, but perhaps most strongly by variation in conflict opportunities. Only once this is accounted for can you make good comparisons in war propensity over time.

With respect to the role of chance, statistical analysis provides the tools you need. The idea is to proceed as if the data you observe (war propensity over time) is the product of both a signal and random background noise. Statistical inference can help quantify how much variation you should expect due to background noise so you can more rigorously judge if a change over time can be easily explained by noise instead of a true signal (or meaningful change).

Adopting a sensible measure of war propensity and applying statistical inference yields results that don't neatly align with the decline of war thesis (at least for one possible measure of war propensity). In short, there's little evidence of a systematic decline in the likelihood of war, at least over the last two centuries.

But the chance of war is only half of the decline of war thesis. Its other major claim is that wars are now less deadly than in the past. Testing this proposition requires wrestling with a host of new conceptual and technical issues, all of which I'll discuss in the next chapter.

References

Braumoeller, Bear F. 2013a. “Is War Disappearing?” In *APSA Chicago 2013 Meeting.*

———. 2013b. *The Great Powers and the International System: Systemic Theory in Empirical Perspective.* Cambridge University Press.

———. 2019. *Only the Dead: The Persistence of War in the Modern Age.* Oxford University Press.

Braumoeller, Bear F, and Austin Carson. 2011. “Political Irrelevance, Democracy, and the Limits of Militarized Conflict.” *Journal of Conflict Resolution* 55 (2): 292–320.

Fearon, James D. 1995. “Rationalist Explanations for War.” *International Organization* 49 (3): 379–414.

Gibler, Douglas M, and Steven V Miller. 2024a. “The Militarized Interstate Confrontation Dataset, 1816-2014.” *Journal of Conflict Resolution* 68 (2-3): 562–86.

———. 2024b. “The Militarized Interstate Events (MIE) Dataset, 1816–2014.” *Conflict Management and Peace Science* 41 (4): 463–81.

Gochman, Charles S, and Zeev Maoz. 1984. “Militarized Interstate Disputes, 1816-1976: Procedures, Patterns, and Insights.” *Journal of Conflict Resolution* 28 (4): 585–616.

Jones, Daniel M, Stuart A Bremer, and J David Singer. 1996. “Militarized Interstate Disputes, 1816–1992: Rationale, Coding Rules, and Empirical Patterns.” *Conflict Management and Peace Science* 15 (2): 163–213.

Mitchell, Sara McLaughlin, and John A Vasquez. 2024. *What Do We Know about War?* Bloomsbury Publishing PLC.

Palmer, Glenn, Roseanne W McManus, Vito D’orazio, Michael R Kenwick, Mikaela Karstens, Chase Bloch, Nick Dietrich, Kayla Kahn, Kellan Ritter, and Michael J Soules. 2022. “The MID5 Dataset, 2011–2014: Procedures, Coding Rules, and Description.” *Conflict Management and Peace Science* 39 (4): 470–82.

Pinker, Steven. 2012. *The Better Angels of Our Nature: Why Violence Has Declined.* Penguin books.

———. 2018. *Enlightenment Now: The Case for Reason, Science, Humanism, and Progress.* Viking.

ReidSarkees, Meredith, and Frank Wayman. 2010. *Resort to War, 1816-2007.* cq Press.

Tuchman, Barbara W. 1994. *The Guns of August: The Outbreak of World War i; Barbara w. Tuchman’s Great War Series.* Random House Trade Paperbacks.

von Clausewitz, Carl. 2003. *On War.* Penguin UK.

Wagner, R Harrison. 2000. “Bargaining and War.” *American Journal of Political Science*, 469–84.